

8. PROOF

Proof of Lemma 3.1. Because $\mathcal{D}_\Pi$ is a compact set, we know that $B = \mathcal{P}_{\mathcal{D}_\Pi}(A)$ exists and is unique. Let $G = U^T B U$, then we have $G \in \mathcal{F}^1$ and $B = U G U^T$. Note that G minimizes $\|A - U G U^T\|_F^2$ over $\mathcal{F}^1$ and

$$\begin{aligned} \|A - U G U^T\|_F^2 &= \|A\|_F^2 - 2\langle A, U G U^T \rangle + \|U G U^T\|_F^2 \\ &= \|A\|_F^2 - 2\langle U^T A U, G \rangle + \|G\|_F^2 \\ &= \|A\|_F^2 - \|U^T A U\|_F^2 + \|U^T A U - G\|_F^2. \end{aligned}$$

Therefore, G is the projection of $U^T A U$ onto $\mathcal{F}^1$ and by Lemma 4.1 of Vu et al. (2013) we have

$$G = \sum_{i=1}^{p-d} \gamma_i^+(\theta) \eta_i \eta_i^T$$

with γ_i , η_i , and θ specified in the theorem. □

Proof of Theorem 3.2. For any given $\tau > 0$, define the augmented Lagrangian of (8) as

$$L_\tau(H, Z, Y) = \mathbb{I}_{\mathcal{D}_\Pi}(H) - \langle S, H \rangle + \rho_2 \|Z\|_{1,1} + \langle Y, H - Z \rangle + \frac{\tau}{2} \|H - Z\|_F^2.$$

The update steps in Algorithm 1 now reads, letting $W^{(r)} = \tau Y^{(r)}$,

$$\begin{aligned} H^{(r)} &= \arg \min_H L_\tau(H, Z^{(r-1)}, W^{(r-1)}), \\ Z^{(r)} &= \arg \min_Z L_\tau(H^{(r)}, Z, W^{(r)}), \\ Y^{(r)} &= Y^{(r-1)} + \tau(H - Z). \end{aligned}$$

It is obvious that $\mathbb{I}_{\mathcal{D}_\Pi}(H) - \langle S, H \rangle$ and $\rho_2 \|Z\|_{1,1}$ are closed, proper, and convex functions. Here we say a function f is closed, proper and convex if $\{(x, t) : f(x) \leq t\}$ is a closed non-empty convex set (Boyd et al. (2011), Section 3.2).

By strong duality, we can find a primal-dual pair of $L_0(H, Z, Y)$, denoted as (H^{**}, Z^{**}, Y^{**}) . It then follows from the primal and dual optimality that (H^{**}, Z^{**}, Y^{**}) is a saddle point of L_0 and hence by Section 3.2.1 of Boyd et al. (2011), we have

$$Z^{(r)} \rightarrow Z^* \quad \text{and} \quad H^{(r)} - Z^{(r)} \rightarrow 0, \quad \text{as } t \rightarrow \infty,$$

where (H^*, Z^*) is an optimal primal variable for L_0 . □

Proof of consistency result. To prove Theorem 4.1, we need some additional lemmas and notation as follows. The proof of lemmas are given after the proof of Theorem 4.1.

Let $I_j = ((j-1)/p, j/p]$ for $j = 2, \dots, p$ and $I_1 = [0, 1/p]$. We define $\phi_j^*(t) = \phi_j(t_i)$ for $t \in I_i$, $u_j^* = p^{-1/2}(\phi_j(t_1), \phi_j(t_2), \dots, \phi_j(t_p))^T$, and $u_j = u_j^*/\|u_j^*\|_2$. Let $\tilde{\Gamma} : [0, 1]^2 \mapsto [0, \infty)$ be such that $\tilde{\Gamma}(s, t) = \Gamma(t_i, t_j)$, if $s \in I_i, t \in I_j$. Define the discretized and diagonal-shifted covariance matrix Σ by $\Sigma(l, l') = \Gamma(t_l, t_{l'}) + a\mathbf{1}(l = l')$. Let $\tilde{\phi}_j$ be eigenfunctions of $\tilde{\Gamma}$ and v_j be eigenvectors of Σ . Then $p^{-1/2}\tilde{\phi}_j(t)$ is the i th entry of v_j if $t \in I_i$. If we further denote the j th eigenvalue of $\tilde{\Gamma}$ by $\tilde{\lambda}_j$, then $(p\tilde{\lambda}_j + a, v_j)$ is an eigenvalue-eigenvector pair of Σ .

Let $\Pi_j = \sum_{i=1}^j v_i v_i^T$, and $\Sigma_j = \Sigma - \Pi_{j-1} \Sigma \Pi_{j-1}$. Define $\Pi_0 = 0$ for convenience. For any measurable $B : [0, 1]^2 \mapsto \mathbb{R}$, let $\|B\|_{\text{HS}} = [\int_{[0,1]^2} B(s, t)^2 ds dt]^{1/2}$ be the Hilbert-Schmidt norm.

Lemma 8.1. *Under Assumptions (A1-A4), let $c_0 = L(2/\delta + 1)$ we have for p large enough, we have $\|v_j - u_j\|_2 \leq c_0 p^{-1}$, for $1 \leq j \leq k$.*

Lemma 8.2. *Under Assumptions (A1-A3), when p is large enough we have $\|\Sigma\|_F^2 \leq c_2^2 p^2$, where $c_2^2 = a^2 + 4\|\Gamma\|_{\text{HS}}^2 + L^2$. Moreover, the gap between the j th and $(j+1)$ th eigenvalues of Σ is at least $p\delta/2$ for all $1 \leq j \leq k$.*

The following lemma is elementary and can be found in Vu and Lei (2013).

Lemma 8.3. *Let u and v be vectors of same length with unit norm. Then*

$$\frac{1}{\sqrt{2}}\|u - v\|_2 \leq \|uu^T - vv^T\|_F \leq \sqrt{2}\|u - v\|_2.$$

The same holds when u, v are functions and $\|\cdot\|_F$ is replaced by $\|\cdot\|_{\text{HS}}$.

The next lemma characterizes u_i as an approximate leading eigenvector of Σ_i . It extends Lemma 4.2 of Vu and Lei (2013).

Lemma 8.4 (Approximate curvature). *Let H_j be a solution to (6). Then under Assumptions (A2-A4), for p large enough,*

$$\frac{p\delta}{8}\|H_j - u_j u_j^T\|_F^2 - \frac{3c_0 c_2}{2} \leq \langle -\Sigma_j, H_j - u_j u_j^T \rangle, \quad \forall \quad 1 \leq j \leq k.$$

Proof of Theorem 4.1. The claim follows if we show that $\sup_{1 \leq j \leq k} \|\widehat{v}_j - u_j\|_2 = o_p(1)$.

For simplicity denote $e_n := \|S - \Sigma\|_{\infty, \infty}$, which is $o_p(1)$ by assumption (A1). Let $\Pi_0 = \widehat{\Pi}_0 = 0$, $\widehat{v}_0 = 0$, $\beta_0 = \epsilon_0 = 0$. We use induction to show that there exist $(\epsilon_i, \beta_i : 0 \leq i \leq k)$ such that

$$\sup_{0 \leq i \leq k} \epsilon_i = o_p(1), \quad \sup_{0 \leq i \leq k} \beta_i = o_p(p), \quad (15)$$

$$\max\{\|\widehat{v}_i - v_i\|_2, \|\widehat{\Pi}_i - \Pi_i\|_2\} \leq \epsilon_i \quad (16)$$

$$\rho_1 \langle D, \widehat{v}_i \widehat{v}_i^T \rangle \leq \beta_i. \quad (17)$$

Obviously the claim holds for $k = 0$. Now assume that the claim holds for $k = j - 1$, and $j \geq 1$. We will construct $\epsilon_j = o_p(1)$ and $\beta_j = o_p(p)$ satisfying (16) and (17).

Let $\Sigma_j = \Sigma - \Pi_{j-1} \Sigma \Pi_{j-1}$ for $j = 1, \dots, k$. By Lemma 8.4 we have, for p large enough,

$$\frac{p\delta}{8}\|H_j - u_j u_j^T\|_F^2 - \frac{3c_0 c_2}{2} \leq \langle -\Sigma_j, H_j - u_j u_j^T \rangle. \quad (18)$$

Next we need to control $\langle \Sigma - \Sigma_j, H_j - u_j u_j^T \rangle$ and $\langle S, H_j - u_j u_j^T \rangle$, where the first one is small because H_j and $u_j u_j^T$ are nearly orthogonal to $\Sigma - \Sigma_j$, and the second term needs to be controlled by the fact that H_j is a maximizer of (6).

For the first term $\langle \Sigma - \Sigma_j, H_j - u_j u_j^T \rangle$, by the orthogonality constraint, we have

$$\langle \Sigma - \Sigma_j, H_j \rangle \leq \lambda_1 \langle \Pi_{j-1}, H_j \rangle = \lambda_1 |\langle \Pi_{j-1} - \widehat{\Pi}_{j-1}, H_j \rangle| \leq \lambda_1 \epsilon_{j-1} \leq c_2 p \epsilon_{j-1}.$$

and similarly

$$\langle \Sigma - \Sigma_j, u_j u_j^T \rangle = \langle \Sigma - \Sigma_j, u_j u_j^T - v_j v_j^T \rangle \leq \|\Sigma_{j-1}\|_F \|u_j u_j^T - v_j v_j^T\|_F \leq \sqrt{2} c_2 c_0,$$

where the last inequality follows from Lemma 8.1 and Lemma 8.3, and therefore

$$|\langle \Sigma - \Sigma_j, H_j - u_j u_j^T \rangle| \leq c_2 p (\epsilon_{j-1} + \sqrt{2} c_0 p^{-1}). \quad (19)$$

Now we turn to the term $\langle S, H_j - u_j u_j^T \rangle$. If we can show that

$$0 \leq \langle S, H_j - u_j u_j^T \rangle - \rho_1 \langle D, H_j \rangle + R_j, \quad (20)$$

for some $R_j = o_p(p)$ then we have, combining (18) to (20),

$$\begin{aligned} \frac{p\delta}{8} \|H_j - u_j u_j^T\|_F^2 &\leq \langle S - \Sigma, H_j - u_j u_j^T \rangle - \rho_1 \langle D, H_j \rangle + R'_j \\ &\leq e_n p \|H_j - u_j u_j^T\|_F - \rho_1 \langle D, H_j \rangle + R'_j, \end{aligned} \quad (21)$$

where $R'_j = c_2 p (\epsilon_{j-1} + \sqrt{2} c_0 p^{-1}) + R_j + 3c_0 c_2 / 2$. It follows that

$$\|H_j - u_j u_j^T\|_F \leq \frac{8e_n}{\delta} + \sqrt{\frac{8R'_j}{\delta p}}. \quad (22)$$

Since $\widehat{v}_j \widehat{v}_j^T$ is the closest rank one, unit norm matrix to H_j , we have

$$\begin{aligned} \|\widehat{v}_j \widehat{v}_j^T - v_j v_j^T\|_F &\leq \|\widehat{v}_j \widehat{v}_j^T - u_j u_j^T\|_F + \|u_j u_j^T - v_j v_j^T\|_F \\ &\leq 2 \|H_j - u_j u_j^T\|_F + \sqrt{2} c_0 p^{-1} \leq \frac{16e_n}{\delta} + 2 \sqrt{\frac{8R'_j}{\delta p}} + \sqrt{2} c_0 p^{-1}, \end{aligned}$$

and

$$\begin{aligned} \|\widehat{\Pi}_j - \Pi_j\|_F &\leq \|\widehat{\Pi}_{j-1} - \Pi_{j-1}\|_F + \|\widehat{v}_j \widehat{v}_j^T - v_j v_j^T\|_F \\ &\leq \epsilon_{j-1} + \frac{16e_n}{\delta} + 2 \sqrt{\frac{8R'_j}{\delta p}} + \sqrt{2} c_0 p^{-1} =: \epsilon_j. \end{aligned}$$

Now it remains to find β_j . Using (21) we have

$$\rho_1 \langle D, H_j \rangle \leq e_n p \|H_j - u_j u_j^T\|_F + R'_j \leq 2e_n p \epsilon_j + R'_j.$$

On the other hand, let $\lambda_{j,1}$ be the largest eigenvalue of H_j . Then

$$\frac{\epsilon_j^2}{4} \geq \|H_j - u_j u_j^T\|_F^2 = \|H_j\|_F^2 - 2\langle H_j, u_j u_j^T \rangle + 1 \geq \lambda_{j,1}^2 - 2\lambda_{j,1} + 1,$$

where we use the fact that $\|H_j\|_F^2 \geq \lambda_{j,1}^2$, and $\langle H_j, u_j u_j^T \rangle \leq \lambda_{j,1} \|u_j\|_2^2$ (von Neumann trace inequality). It then follows that $\lambda_{j,1} \geq 1 - \epsilon_j/2$, which implies that

$$\rho_1 \langle D, \widehat{v}_j \widehat{v}_j^T \rangle \leq (1 - \epsilon_j/2)^{-1} \rho_1 \langle D, H_j \rangle \leq (1 - \epsilon_j/2)^{-1} (2e_n p \epsilon_j + R'_j) =: \beta_j.$$

Direct verification shows that if $\max_{0 \leq i \leq j-1} \epsilon_i = o_p(1)$, $\max_{0 \leq i \leq j-1} \beta_i = o_p(p)$, and $R_j = o_p(p)$, then $\epsilon_j = o_p(1)$ and $\beta_j = o_p(p)$.

The rest of the proof is to show (20) for some $R_j = o_p(p)$. The main challenge is that u_j is not in the feasible set of problem (6) and hence $u_j u_j^T$ is not directly comparable to H_j using optimality condition of (6). To overcome this difficulty, we consider $\tilde{u}_j$, a modified version of u_j so that (a) $\tilde{u}_j$ is close to u_j in ℓ_2 norm; (b) $\tilde{u}_j \tilde{u}_j^T$ is feasible for (6); (c) $\tilde{u}_j$ is almost as smooth as u_j .

Define $\tilde{u}_j = (I - \widehat{\Pi}_{j-1})u_j / \|(I - \widehat{\Pi}_{j-1})u_j\|$. We first check the validity of this definition.

$$\|\widehat{\Pi}_{j-1} u_j\|_2 \leq \|(\widehat{\Pi}_{j-1} - \Pi_{j-1})u_j\|_2 + \|\Pi_{j-1} u_j\|_2 + \|\Pi_{j-1}(u_j - v_j)\|_2 \leq \epsilon_{j-1} + c_0 p^{-1}.$$

When ϵ_{j-1} is small and p large, $(I - \widehat{\Pi}_{j-1})u_j \neq 0$, and

$$\|\tilde{u}_j - u_j\|_2 = \left\| \frac{(I - \widehat{\Pi}_{j-1})u_j}{\|(I - \widehat{\Pi}_{j-1})u_j\|_2} - \frac{u_j}{\|u_j\|_2} \right\|_2 \leq 2\|(I - \widehat{\Pi}_{j-1})u_j - u_j\|_2 \leq 2(\epsilon_{j-1} + c_0 p^{-1}), \quad (23)$$

where the last inequality holds when p is large and the first inequality follows from an elementary fact that, for all u, v ,

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 \leq 2 \frac{\|u - v\|_2}{\max(\|u\|_2, \|v\|_2)}.$$

Now we establish (20). By feasibility of $\tilde{u}_j \tilde{u}_j^T$ we have

$$\begin{aligned} 0 &\leq \langle S, H_j - \tilde{u}_j \tilde{u}_j^T \rangle - \rho_1 \langle D, H_j \rangle + \rho_1 \langle D, \tilde{u}_j \tilde{u}_j^T \rangle - \rho_2 (\|H_j\|_{1,1} - \|\tilde{u}_j \tilde{u}_j^T\|_{1,1}) \\ &\leq \langle S, H_j - u_j u_j^T \rangle - \rho_1 \langle D, H_j \rangle + \rho_1 \langle D, \tilde{u}_j \tilde{u}_j^T \rangle + \rho_2 p + |\langle S, \tilde{u}_j \tilde{u}_j^T - u_j u_j^T \rangle|. \end{aligned} \quad (24)$$

We first bound $|\langle S, \tilde{u}_j \tilde{u}_j^T - u_j u_j^T \rangle|$:

$$\begin{aligned} |\langle S, \tilde{u}_j \tilde{u}_j^T - u_j u_j^T \rangle| &\leq \|S\|_F \|\tilde{u}_j \tilde{u}_j^T - u_j u_j^T\|_F \leq (\|\Sigma\|_F + \|S - \Sigma\|_F) \|\tilde{u}_j \tilde{u}_j^T - u_j u_j^T\|_F \\ &\leq 2\sqrt{2} (c_2 + e_n) p (\epsilon_j + c_0 p^{-1}), \end{aligned}$$

where the last step uses (23), Lemma 8.2, and the fact that $\|S - \Sigma\|_{\infty, \infty} = e_n$.

Now we control $\rho_1 \langle D, \tilde{u}_j \tilde{u}_j^T \rangle$. When ϵ_{j-1} and p^{-1} are small enough such that $\|(I - \hat{\Pi}_{j-1})u_j\|_2 \geq 1/\sqrt{2}$, we have

$$\begin{aligned} \rho_1 \langle D, \tilde{u}_j \tilde{u}_j^T \rangle &= \rho_1 \|\Delta \tilde{u}_j\|_2^2 \leq 2\rho_1 \|\Delta(I - \hat{\Pi}_{j-1})u_j\|_2^2 \\ &\leq 4\rho_1 \left[\|\Delta u_j\|_2^2 + \left(\sum_{i=1}^{j-1} \|\Delta \hat{v}_i\|_2 |\langle \hat{v}_i, u_j \rangle| \right)^2 \right] \leq 4 \left[2\rho_1 L^2 p^{-4} + \sum_{i=1}^{j-1} \beta_i (\epsilon_{j-1} + c_0 p^{-1})^2 \right], \end{aligned}$$

where the first two inequalities follow from multiple applications of Cauchy-Schwartz, and the last inequality holds by definition of β_i , the smoothness of u_j , and the fact that $\sum_{i=1}^{j-1} |\langle \hat{v}_i, u_j \rangle|^2 = \|\hat{\Pi}_{j-1} u_j\|_2^2$. As a consequence, we establish (20) from (24) with

$$R_j = 2\sqrt{2} (c_2 + e_n) (\epsilon_{j-1} p + c_0) + \rho_2 p + 8 \left[\rho_1 L^2 p^{-4} + \sum_{i=1}^{j-1} \beta_i (\epsilon_{j-1} + c_0 p^{-1})^2 \right] = o_p(p). \quad \square$$

Proof of Theorem 4.2. From assumption A1 and Lemma 8.1 it suffices to prove that if $\|S - \Sigma\|_{\infty, \infty} = O(e_n)$ then $\sup_{1 \leq j \leq k} \|\hat{v}_j - v_j\|_2 = O(e_n + \rho_2)$.

Consider the estimation procedure given by (5) for $j = 1$. Let $\mathbb{B}_1$ be the collection of all $p \times p$ symmetric matrices with entries in $[-1, 1]$. The optimization problem can be written in the following equivalent form.

$$\max_{H \in \mathcal{F}^1} \min_{Z \in \mathbb{B}_1} \langle S, H \rangle - \rho_2 \langle Z, H \rangle.$$

Let H^* be any maximizer, then

$$H^* = \arg \max_{H \in \mathcal{F}^1} \langle S - \rho_2 Z^*, H \rangle = \arg \max_{H \in \mathcal{F}^1} \langle \Sigma + W - \rho_2 Z^*, H \rangle$$

where $W = S - \Sigma$ and Z^* is the corresponding optimal dual variable.

By Lemma 8.1 the eigengap of Σ is of order at least p while the operator norm of $W - \rho_2 Z^*$ is at most $p(\|W\|_{\infty, \infty} + \rho_2)$ which is $o(p)$. Thus applying standard spectral subspace perturbation theory we know that $H^* = \hat{v}_1 \hat{v}_1^T$ where $\hat{v}_1$ is the leading eigenvector of $\Sigma + W - \rho_2 Z^*$, and satisfies for some constant c

$$\|\hat{v}_1 - v_1\|_2 \leq c\|W - \rho_2 Z^*\|_F/p \leq O(e_n + \rho_2).$$

For $j = 2, \dots, k$, we use induction. Suppose that for $j-1$ we have $\|\hat{v}_{j-1} - v_{j-1}\|_2$ and $\|\hat{\Pi}_{j-1} - \Pi_{j-1}\|_F$ are bounded by $O(e_n + \rho_2)$.

Now consider the procedure (5) for j . Similarly let H^* be any solution and Z^* the corresponding optimal dual variable. We have

$$\begin{aligned} H^* &= \arg \max_{H \in \mathcal{D}_{\hat{\Pi}_{j-1}}} \langle S - \rho_2 Z^*, H \rangle \\ &= \arg \max_{H \in \mathcal{F}^1} \langle (I - \hat{\Pi}_{j-1})(S - \rho_2 Z^*)(I - \hat{\Pi}_{j-1}), H \rangle. \end{aligned}$$

The remainder of the proof focuses on analyzing the matrix $(I - \hat{\Pi}_{j-1})(S - \rho_2 Z^*)(I - \hat{\Pi}_{j-1})$. In particular, we show that its leading eigenvector is close to v_j with the desired rate. We first write this matrix in four terms

$$\begin{aligned} &(I - \hat{\Pi}_{j-1})(\Sigma + W - \rho_2 Z^*)(I - \hat{\Pi}_{j-1}) \\ &= (I - \Pi_{j-1})\Sigma(I - \Pi_{j-1}) \\ &\quad + (\Pi_{j-1} - \hat{\Pi}_{j-1})\Sigma(I - \hat{\Pi}_{j-1}) + (I - \Pi_{j-1})\Sigma(\Pi_{j-1} - \hat{\Pi}_{j-1}) \\ &\quad + (I - \hat{\Pi}_{j-1})(W + \rho_2 Z^*)(I - \hat{\Pi}_{j-1}) \end{aligned}$$

$$=T_0 + T_1 + T_2.$$

The main term is T_0 . The leading eigenvector of T_0 is v_j with an eigengap of order at least p . Next we bound the operator norms of T_1 and T_2 . In fact we have

$$\|T_1\|_F \leq \|(\Pi_{j-1} - \widehat{\Pi}_{j-1})\Sigma\|_F + \|\Sigma(\Pi_{j-1} - \widehat{\Pi}_{j-1})\|_F \leq 2\|\Pi_{j-1} - \widehat{\Pi}_{j-1}\|_F \|\Sigma\|_2 = O((e_n + \rho_2)p),$$

and

$$\|T_2\|_F \leq \|W + \rho_2 Z^*\|_F \leq (e_n + \rho_2)p.$$

The induction is complete by applying the spectral perturbation result again. $\square$

Proof of Technical Lemmas

Proof of Lemma 8.1. Note that $\widetilde{\Gamma}$ is a compact self-adjoint operator from $L^2(0, 1)$ to $L^2(0, 1)$ with eigen-decomposition $\widetilde{\Gamma}(s, t) = \sum_{j=1}^p \widetilde{\lambda}_j \widetilde{\phi}_j(s) \widetilde{\phi}_j(t)$.

The Lipschitz condition on Γ implies that

$$\|\widetilde{\Gamma} - \Gamma\|_{\text{HS}}^2 := \int \int |\Gamma_p(s, t) - \Gamma(s, t)|^2 ds dt \leq \frac{L^2}{4p^2}. \quad (25)$$

By Weyl's inequality, $|\widetilde{\lambda}_j - \lambda_j| \leq \delta/2$ for large p . Let E_j and $\widetilde{E}_j$ be the projection operators onto the one-dimensional subspaces spanned by ϕ_j and $\widetilde{\phi}_j$, respectively. Then

$$\|\widetilde{\phi}_j - \phi_j\|_2 \leq \sqrt{2} \|\widetilde{E}_j - E_j\|_{\text{HS}} \leq \frac{4\|\widetilde{\Gamma} - \Gamma\|_{\text{HS}}}{\delta} \leq \frac{2L}{\delta p}, \quad (26)$$

where the first inequality follows from Lemma 8.3, and the second inequality follows from the Davis-Kahan $\sin \Theta$ theorem (Chapter VII of Bhatia (1997)).

On the other hand, by assumption (A4) we have

$$\|\phi_j^* - \phi_j\|_2 \leq \|\phi_j^* - \phi_j\|_\infty \leq \frac{L}{2p}. \quad (27)$$

which, together with (26), implies that

$$\|v_j - u_j^*\|_2 \leq \left(\frac{2}{\delta} + \frac{1}{2}\right) \frac{L}{p}.$$

Also note that

$$\|u_j^* - u_j\|_2 = \left| \|u_j^*\|_2 - 1 \right| \leq \|\phi_j^* - \phi_j\|_2 \leq \|\phi_j^* - \phi_j\|_\infty \leq \frac{L}{2p}.$$

Combining the previous two inequalities, we have

$$\|v_j - u_j\|_2 \leq \left(\frac{2}{\delta} + 1\right) \frac{L}{p} := c_0 p^{-1}. \quad \square$$

Proof of Lemma 8.2. The first claim follows from, letting Σ^* be the discretized Γ evaluated at the grid,

$$\begin{aligned} \|\Sigma\|_F^2 &\leq 2\|aI_p\|_F^2 + 2\|\Sigma^*\|_F^2 = 2a^2p + 2p^2\|\tilde{\Gamma}\|_{\text{HS}}^2 \\ &\leq 2a^2p + 2p^2 \left(2\|\Gamma\|_{\text{HS}}^2 + 2\|\tilde{\Gamma} - \Gamma\|_{\text{HS}}^2 \right) \leq 2a^2p + 2p^2 \left(2\|\Gamma\|_{\text{HS}}^2 + \frac{L^2}{2p^2} \right) \\ &\leq p^2 (2a^2p^{-1} + 4\|\Gamma\|_{\text{HS}}^2 + L^2) \leq c_2^2 p^2, \end{aligned}$$

where (25) is used to bound $\|\tilde{\Gamma} - \Gamma\|_{\text{HS}}$.

The second claim follows from the fact that the eigengaps of Σ are the same as those of Σ^* , and by Weyl's inequality:

$$\tilde{\lambda}_j - \tilde{\lambda}_{j+1} \geq \lambda_j - \lambda_{j+1} - 2\|\tilde{\Gamma} - \Gamma\|_{\text{HS}} \geq \delta - \frac{\delta}{2} = \delta/2. \quad \square$$

Proof of Lemma 8.4. Note that v_j is the leading eigenvector of Σ_j , with eigengap at least $p\delta/2$ as implied by Lemma 8.2. Then we have

$$\begin{aligned} \|H_j - u_j u_j^T\|_F^2 &\leq 2\|H_j - v_j v_j^T\|_F^2 + 2\|v_j v_j^T - u_j u_j^T\|_F^2 \leq \frac{8}{p\delta} \langle -\Sigma, H_j - v_j v_j^T \rangle + 4c_0^2 p^{-2} \\ &\leq \frac{8}{p\delta} \langle -\Sigma, H_j - u_j u_j^T \rangle + \frac{8}{p\delta} \|\Sigma\|_F \|u_j u_j^T - v_j v_j^T\|_F + 4c_0^2 p^{-2} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{8}{p\delta} \langle -\Sigma, H_j - u_j u_j^T \rangle + \frac{8\sqrt{2}c_0 c_2}{p\delta} + 4c_0^2 p^{-2} \\
&\leq \frac{8}{p\delta} \langle -\Sigma, H_j - u_j u_j^T \rangle + \frac{12c_0 c_2}{p\delta},
\end{aligned}$$

where the first and third inequalities come from Cauchy-Schwartz, the second from the curvature lemma of principal subspace (Vu and Lei (2013), Lemma 4.2), and the last holds provided that p is sufficiently large. $\square$